

Factory Method Pattern Example

```
interface Document {  
    void open();  
}  
  
// Word document  
class WordDocument implements Document {  
    public void open() {  
        System.out.println("Word Document created");  
    }  
}  
  
// Pdf document  
class PdfDocument implements Document {  
    public void open() {  
        System.out.println("PDF Document created");  
    }  
}  
  
// Excel document  
class ExcelDocument implements Document {  
    public void open() {  
        System.out.println("Excel Document created");  
    }  
}
```

```
abstract class DocumentFactory {  
    abstract Document createDocument();  
}
```

```
class WordDocument implements Document {  
    public void open() {  
        System.out.println("Word Document created");  
    }  
}
```

```
class PdfFactory extends DocumentFactory {  
    Document createDocument() {  
        return new PdfDocument();  
    }  
}
```

```
class ExcelFactory extends DocumentFactory {  
    Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

```
#Main.java

public class FactoryMethodTest {

    public static void main(String[] args) {

        // Word Document

        DocumentFactory wordFactory = new WordFactory();

        Document word = wordFactory.createDocument();

        word.open();

        // PDF Document

        DocumentFactory pdfFactory = new PdfFactory();

        Document pdf = pdfFactory.createDocument();

        pdf.open();

        // Excel Document

        DocumentFactory excelFactory = new ExcelFactory();

        Document excel = excelFactory.createDocument();

        excel.open();

    }

}
```